

UNDERSTANDING RELATIVITY

EINSTEIN'S GROUND-BREAKING THEORIES REVEALED THAT SPACE AND TIME ARE INTIMATELY LINKED

In the early 20th century, German-born physicist Albert Einstein published two theories that revolutionized our understanding of space, time, energy, and gravity. The first, known as the special theory of relativity, only applies in certain circumstances; the second is the general theory of relativity.

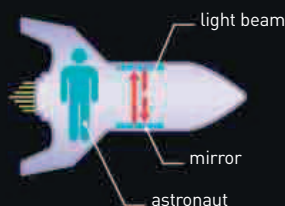
In the 19th century, physicists thought that "empty" space was actually filled with a substance, which they called ether, and that light travels through the ether at a fixed speed. Since our planet is moving, they predicted that the measured speed of light would differ from its actual speed—just as a passing car appears to move faster or slower than its actual speed if you are also moving. To test this idea—and in an attempt to determine Earth's actual, or "absolute" speed through space—they measured the speed of light in different directions and at different times of the year. But in every case, the speed of light was always exactly the same.

SPECIAL THEORY OF RELATIVITY

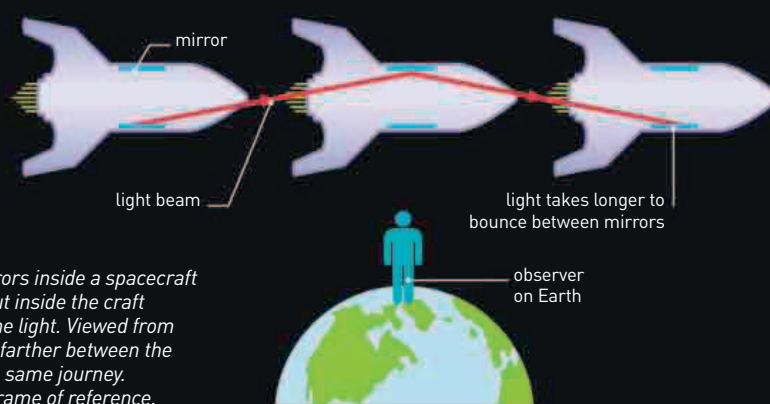
The fact that the speed of light is constant was perplexing, and it challenged common-sense assumptions about the nature of time and space. In his theory of special relativity Einstein showed that time and space are indeed "relative" quantities: that two observers in relative motion

both measuring the distance between the same two points in space, or the time between the same two events, would come up with different answers. Einstein's relativity also did away with the "stationary" ether and showed that there is no absolute reference point in space or time.

OBSERVER WITHIN REFERENCE FRAME

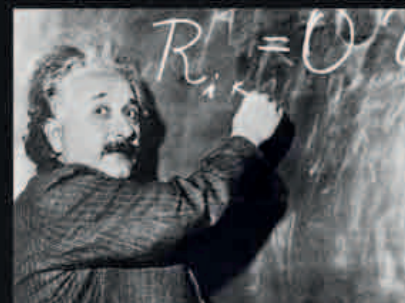


OBSERVER OUTSIDE REFERENCE FRAME



TIME DILATION

A beam of light bounces between mirrors inside a spacecraft that is moving past Earth. An astronaut inside the craft perceives only vertical movement in the light. Viewed from Earth, it appears that the light travels farther between the mirrors, and takes longer to make the same journey. So, time runs slowly in the "moving" frame of reference.



ALBERT EINSTEIN

When Einstein published his theory of special relativity, he was working as a clerk at the patent office in Bern, Switzerland.

“IT FOLLOWED FROM THE **SPECIAL THEORY OF RELATIVITY** THAT **MASS AND ENERGY** ARE... MANIFESTATIONS OF **THE SAME THING.**”

Albert Einstein, in the film *Atomic Physics*, 1948

MASS AND ENERGY

In working through the mathematical equations in his theory of relativity, Einstein encountered a surprising result: the mass of an object increases as it gets faster, and at the speed of light an object would have infinite mass. Einstein realized that the speed of light must therefore be the Universe's ultimate speed limit. His equations suggested that mass and energy are equivalent to each other, and he defined a new quantity: "mass-energy." The equivalence of mass and energy is expressed in Einstein's most famous equation, $E = mc^2$, in which E stands for energy, m for mass, and c^2 for the speed of light squared.



$$E = mc^2$$

energy mass speed of light

PARTICLE ACCELERATOR

Physicists working with particle accelerators routinely use Einstein's theory to predict the increase in mass of high-speed particles, and to work out how much longer it takes for them to decay as a result of time dilation (see above).